Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

***Benchmark:** AFRIDOC_MT* | ***Context:** Amhara Region Agricultural & Microfinance Document Translation — Cooperative and Rural MFI End-Users*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is the most aligned dimension and the user assigned it LOWER priority. Both benchmark and deployment are text-only, and Amharic is explicitly handled in Ethiopic (Ge'ez) script with documented accommodations: token limits were specifically increased for Amharic [Q162] because its non-Latin script produced larger token counts than English [Q127]. The benchmark adopts pseudo-document chunking with k=10 to fit Amharic within model context limits [Q47, Q48, Q128], a practical choice that the deployment will likely also need. Empirically, all sampled Amharic content is rendered consistently in Ethiopic script across all configurations [DATASET-D1, DATASET-D5, Strength 1]. The remaining concern is that the deployment may evolve toward format-preserving or multimodal translation (scanned PDFs, structured documents with tables and signature blocks) — a possibility flagged but unconfirmed in the regional context. Some annotation-workflow infrastructure (CSV/Google Sheets with paragraph-marking empty rows) [Q119, Q122] partially preserves document structure at the corpus level. Score of 4 reflects strong technical alignment on script and tokenization, with a minor gap on structural fidelity that becomes relevant only if deployment expands to format-preserving use cases.

ID	Evidence
Q47	"An initial analysis showed that some models were unable to process entire documents due to input length limits, which were exceeded by token counts in some languages (Amharic and Yorùbá)." (p.5)
Q48	"To address this, we adopted a similar approach to Lee et al. (2022), splitting documents into fixed-size chunks of k sentences to fit within token limits; the final chunk may contain fewer than k sentences." (p.5)
Q119	"Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text." (p.18)
Q122	"The files are in .csv format, and you can open them using Google Sheets or Microsoft Excel (for offline work)." (p.18)
Q127	"Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers." (p.19)
Q128	"To accommodate both languages in our experiments, we chose pseudo-documents with k=10." (p.19)
Q162	"Therefore, we increased the token limit specifically for Amharic." (p.21)
D1	ሞቃታማ አውሎ ነፋሶችን እንዲሁም ሀይለኛ አውሎ ነፋስ ወይም አውሎ ነፋሶች በመባል የሚታወቁት በጣም አጥፊ የአየር ሁኔታ ክስተቶች ናቸው — Amharic translation of WHO health content on tropical cyclones — health domain, Ethiopic script
D5	ይህ የሚያሳየው የደብሊውፒቪ1 ስርጭት ከአፍጋኒስታን (ምስራቅ ክልል) እና ፓኪስታን (ደቡብ ኬፒ) ዞኖች — Amharic: polio surveillance language, mentions sub-national zones and infection control measures

Strengths

- Amharic Ethiopic script is correctly handled across all configurations with explicit token-limit accommodation [Q162, DATASET-D1]
- Multiple chunk-size configurations (k=5, 10, 25, full document, sentence-level) [Q49, DATASET-D5] enable matching deployment document lengths
- Annotation workflow preserves paragraph-separating empty rows at corpus level [Q119]
- Token-length statistics across languages [Q33] demonstrate empirical engagement with Amharic morphological characteristics

ID	Evidence
Q33	“Table 4 shows the average number of whitespace-separated tokens per sentence across domains and languages, including English. The health domain has more tokens than tech. Hausa and Yorùbá are longer than English, likely due to their descriptive nature, while Swahili is similar in length. Amharic and Zulu are relatively shorter, reflecting interesting linguistic properties.” (p.4)
Q49	“To select an appropriate chunk size, we conducted initial tests with k = 1 (sentence-level), 5, 10, and 25, choosing k = 10 for our experiments.” (p.5)
Q119	“Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text.” (p.18)
Q162	“Therefore, we increased the token limit specifically for Amharic.” (p.21)
D1	ሞቃታማ አውሎ ነፋሶችን እንዲሁም ሀይለኛ አውሎ ነፋስ ወይም አውሎ ነፋሶች በመባል የሚታወቁት በጣም አጥፊ የአየር ሁኔታ ክስተቶች ናቸው — Amharic translation of WHO health content on tropical cyclones — health domain, Ethiopic script
D5	ይህ የሚያሳየው የደብሊውፒቪ1 ስርጭት ከአፍጋኒስታን (ምስራቅ ክልል) እና ፓኪስታን (ደቡብ ኬፒ) ዞኖች — Amharic: polio surveillance language, mentions sub-national zones and infection control measures

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution match is good. Both benchmark and deployment are text-only Amharic in Ethiopic script. Deployment delivery channel is print/SMS/WhatsApp [regional context: technology_access], all of which are text representations the benchmark covers.

IF-2: Amharic Ethiopic-script text infrastructure is supported in the benchmark; rural Amhara connectivity issues affect distribution but not the input form the MT system processes [WEB-18 on internet blockages affects delivery, not signal format].

IF-3: Domain-specific form differences: (a) deployment may include scanned PDFs requiring OCR [regional context: multimodal_future NEEDS VERIFICATION] — not assessed by AFRIDOC-MT; (b) deployment documents may include numbered clause structures, tables, signature blocks — partially preserved at corpus level [Q119] but no metric assesses preservation [Q57]; (c) Amharic register-specific token patterns of formal proclamation Amharic may differ from health/IT news Amharic — not directly addressed.

IF-4: Form mismatches harming external validity: minor — text-only modality matches; structural fidelity gap is the only documented mismatch and is contingent on multimodal evolution. INSUFFICIENT DOCUMENTATION on whether deployment receives plain text vs. structured documents; would need consultation with Amhara Regional Bureau.

ID	Evidence
Q33	“Table 4 shows the average number of whitespace-separated tokens per sentence across domains and languages, including English. The health domain has more tokens than tech. Hausa and Yorùbá are longer than English, likely due to their descriptive nature, while Swahili is similar in length. Amharic and Zulu are relatively shorter, reflecting interesting linguistic properties.” (p.4)
Q47	“An initial analysis showed that some models were unable to process entire documents due to input length limits, which were exceeded by token counts in some languages (Amharic and Yorùbá).” (p.5)
Q57	“In Tables 5 and 6 we present d-chrF scores based on the realigned documents, created by merging the translated sentences into their corresponding documents.” (p.5)
Q119	“Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text.” (p.18)
Q127	“Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers.” (p.19)

Q162	“Therefore, we increased the token limit specifically for Amharic.” (p.21)
WEB-18	Internet blockages in Amhara region affect document delivery channels but not text-input signal format (https://www.aljazeera.com/features/2025/3/10/we-just-want-peace-a-pacifist-community-amid-ethiopias-amhara-conflict)

Information Gaps

- Whether deployment receives documents as plain text, structured digital files, or scanned PDFs requiring OCR
- Per-token-count statistics for Amharic proclamation register vs. AFRIDOC-MT health Amharic

Requires Expert Verification

- Empirical token-length comparison between AFRIDOC-MT Amharic and bureau proclamation Amharic to assess whether the chosen chunk size generalizes